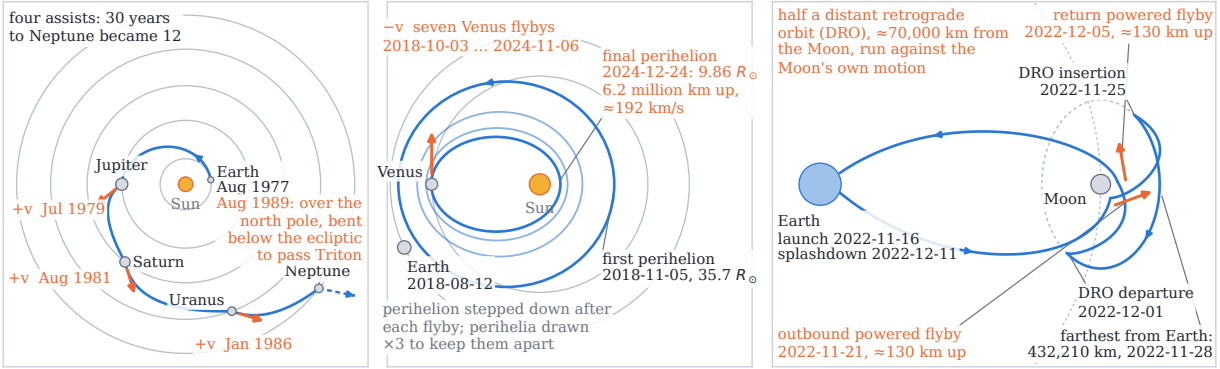
(a) Voyager 2
four assists: 30 years
to Neptune became 12
Jupiter
Earth
Aug 1977
Sun
+v Jul 1979
Aug 1989: over the
north pole, bent
below the ecliptic
to pass Triton
Saturn
+v Aug 1981
Neptune
Uranus
+v Jan 1986
(b) Parker Solar Probe
−v seven Venus flybys
2018-10-03 … 2024-11-06
final perihelion
2024-12-24: 9.86 $R_\odot$
6.2 million km up,
≈192 km/s
Venus
Sun
Earth
2018-08-12
first perihelion
2018-11-05, 35.7 $R_\odot$
perihelion stepped down after
each flyby; perihelia drawn
×3 to keep them apart
(c) Artemis I, rotating frame
half a distant retrograde
orbit (DRO), ≈70,000 km from
the Moon, run against the
Moon's own motion
return powered flyby
2022-12-05, ≈130 km up
DRO insertion
2022-11-25
Moon
Earth
launch 2022-11-16
splashdown 2022-12-11
DRO departure
2022-12-01
outbound powered flyby
2022-11-21, ≈130 km up
farthest from Earth:
432,210 km, 2022-11-28